

DATA STRUCTURES LAB

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5CS04	ESC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OBJECTIVES

The course should enable the students to:

1. Ability to identify the appropriate data structure for given problem.
2. Effectively use compilers include library functions, debuggers and trouble shooting.
3. Write and execute programs using data structures such as arrays, linked lists to implement stacks, queues.
4. Write and execute programs in C to implement various sorting and searching.

COURSE OUTCOMES

The course should enable the students to:

1. Use appropriate data structure for given problem.
2. Use compilers include library functions, debuggers and trouble shooting.
3. Execute write programs in C to implement various types Linked Lists.
4. Execute programs using data structures such as arrays, linked lists to implement stacks.
5. Execute programs using data structures such as arrays, linked lists to implement queues.
6. Execute write programs in C to implement various sorting and searching.

LIST OF EXPERIMENTS

WEEK - 1

STRUTCURES

Write a C Program using functions to

- a. Reading a complex number
- b. Writing a complex number
- c. Add two complex numbers
- d. Multiply two complex numbers

Note: represent complex number using structure.

WEEK - 2

ARRAYS

- a. Write a C program
 - iii. To add two matrices
 - iv. To multiply two matrices
- b. Write a C program to implement Sparse Matrices.

WEEK - 3

SINGLE LINKED LIST

Write a C program that uses functions to perform the following:

- a. Create a singly linked list of integers.
- b. Delete a given integer from the above linked list.
- c. Display the contents of the above list after deletion.

WEEK - 4

SINGLE LINKED LIST

Write a C program that uses functions to perform the following: a. Create TWO singly linked list of integers. b. Concatenate TWO Singly Linked Lists. c. Display the contents of the above list after concatenation.	
WEEK - 5	DOUBLE LINKED LIST
Write a C program that uses functions to perform the following: a. Create a doubly linked list of integers. b. Delete a given integer from the above doubly linked list. c. c) Display the contents of the above list after deletion.	
WEEK - 6	STACK
Write C programs to implement a Queue ADT using i) array ii) linked list	
WEEK - 7	STACK APPLICATION
a. Write a C program that uses stack operations to convert a given infix expression into its postfix Equivalent, Implement the stack using an array. b. Write a C program that uses Stack to evaluate Postfix Expression.	
WEEK - 8	QUEUE
Write C programs to implement a Queue ADT using i) array ii) linked list	
WEEK - 9	DOUBLE ENDED QUEUE
Write C programs to implement a double ended queue ADT using i) array ii) doubly linked list	
WEEK - 10	SEARCHING
Write C programs for implementing the following searching methods: a) Linear Search b) Binary Search	
WEEK - 11	SORTING
Write C programs for implementing the following sorting methods to arrange a list of integers in Ascending order : a) Insertion sort b) Merge sort	
WEEK - 12	SORTING
Write C programs for implementing the following sorting methods to arrange a list of integers in ascending order: a) Quick sort b) Selection sort	
TEXT BOOKS	

1. C and Data Structures, Prof. P.S.Deshpande and Prof. O.G. Kakde, Dreamtech Press.
2. Data structures using C, A.K.Sharma, 2nd edition, Pearson.
3. Data Structures using C, R.Thareja, Oxford University Press.

WEB REFERENCES

1. <http://www.sanfoundry.com/data-structures-examples>
2. <http://www.geeksforgeeks.org/c>
3. <http://www.cs.princeton.edu>